

Pablo

Calcular $\|T\|$

$$T: (C[0,1], \|\cdot\|_\infty) \rightarrow (C[0,1], \|\cdot\|_\infty)$$

$$f(x) \mapsto x^2 \cdot f(x)$$

1) Lineal

2) Calcular $\|T\|$

$$T(\lambda f + g) = \lambda T(f) + T(g)$$

$$\begin{aligned} x^2(\lambda f + g)(x) &= x^2 \cdot (\lambda f(x) + g(x)) = \lambda x^2 f(x) + x^2 g(x) \\ &= \lambda T(f) + T(g) \end{aligned}$$

$$\|T\| = \sup_{f \in B(0,1)} \|T(f)\|_\infty$$

$$\|Tf\|_\infty = \sup_{x \in [0,1]} |x^2 f(x)|$$

$$f \in B(0,1)$$

$$\|f\|_\infty \leq 1$$

$$\sup_{x \in [0,1]} |f(x)| \leq 1$$

$$\sup_{x \in [0,1]} |x^2 f(x)| \leq |x^2| |f(x)| \leq \underbrace{|x^2|}_{\leq 1} \cdot \|f\|_\infty \leq \|f\|_\infty$$

$$\|Tf\|_\infty \leq \|f\|_\infty$$

$$\sup_{f \in B(0,1)} \|Tf\|_\infty \leq 1$$

$$\Rightarrow \|T\| \leq 1$$

$$\underbrace{f \in B(0,1)}_{\|f\|_\infty \leq 1}$$

$$\text{Sup } \exists f \in B(0,1) / \|Tf\| = 1 \Rightarrow \|T\| = 1$$

$$\text{Si } f=1 \quad \|Tf\| = \sup_{x \in [0,1]} |x^2 f(x)| = \sup_{x \in [0,1]} |x^2| = 1$$

$$\boxed{\begin{aligned} (f_n) &\subseteq B(0,1) \\ \|Tf_n\| &\rightarrow 1 \end{aligned}}$$

$$\Rightarrow \|T\| = 1$$

$$T: (C[0,1], \|\cdot\|_\infty) \rightarrow (C[0,1], \|\cdot\|_\infty)$$

$$Tf(x) = \int_0^x (x-t)f(t)dt$$

1) T es lineal ✓

$$\|Tf\|_\infty = \sup_{x \in [0,1]} \left| \int_0^x (x-t)f(t)dt \right|$$

$$\left| \int_0^x (x-t)f(t)dt \right| \leq \int_0^x |x-t| |f(t)| dt \leq \|f\|_\infty \int_0^x \underbrace{|x-t|}_{\geq 0} dt =$$

$$\begin{array}{c} |f(t)| \leq |f(t)| \leq \|f\|_\infty \\ \tau \in [0,x] \quad \tau \in [0,1] \end{array}$$

$$= \|f\|_\infty \int_0^x x-t dt = \|f\|_\infty \left(x\tau - \frac{\tau^2}{2} \right) \Big|_0^x = \|f\|_\infty \left(x^2 - \frac{x^2}{2} \right) = \|f\|_\infty \frac{x^2}{2}$$

$$\begin{aligned} \sup_{x \in [0,1]} |Tf(x)| &\leq \sup_{x \in [0,1]} \|f\|_\infty \frac{x^2}{2} \\ &\quad \downarrow \\ \|Tf\|_\infty &\leq \frac{\|f\|_\infty}{2} \end{aligned}$$

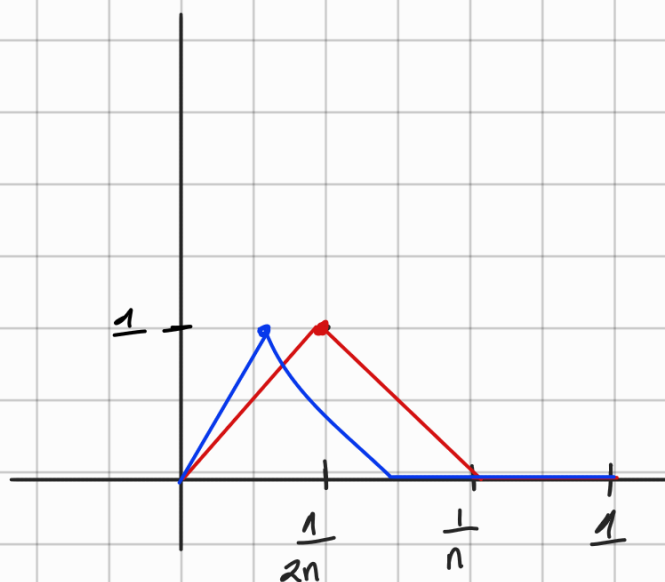
$$\begin{array}{l} \sup_{f \in B(0,1)} \|Tf\|_\infty \leq \frac{1}{2} \end{array}$$

$$\Rightarrow \|T\| \leq \frac{1}{2}$$

Sea $f \equiv 1$. Calculo $\|Tf\|_{\infty} = \sup_{x \in [0,1]} \left| \int_0^x (x-\tau) d\tau \right|$

Sucesiones de funciones

$$f_n(x) = \begin{cases} 0 & \text{si } x \geq 1/n \\ nx^2 & \text{si } x \leq 1/2n \\ -n(x - \frac{1}{n}) & \text{si } \frac{1}{2n} < x < \frac{1}{n} \end{cases}$$



Sea $x \in [0,1]$

$$x=0 \Rightarrow f_n(0) = 0 \quad \forall n \in \mathbb{N}$$

$$f_n(0) \xrightarrow{n} 0$$

$$\Rightarrow f_n(x) \xrightarrow{\text{puntualmente}} f(x) \equiv 0$$

$$f_n \not\Rightarrow f \text{ si dado } \varepsilon > 0, \exists n_0 / d(f_n(x), f(x)) < \varepsilon \quad \forall x$$

Queda como Tarea ver la
convergencia uniforme, ¡spoiler! $\rightarrow$ no